

Title

Analyzing Factors Influencing Food Delivery Time

Executive Summary

This analysis examines how delivery personnel characteristics and external conditions influence delivery time using a dataset of over 45,000 delivery transactions across urban and metropolitan areas in India.

The findings found that the workload is the strongest operational driver of delivery delays. The number of multiple deliveries shows the highest correlation with delivery time ($r=0.387$), meaning couriers handling more than one order simultaneously face measurably longer delivery times. This represents a direct, internal factor of inefficiency.

Courier ratings are a substantial predictor of performance. Higher-rated delivery personnel complete orders faster, with a moderate negative correlation ($r= -0.339$), suggesting that experience and professionalism translate into tangible time savings. In contrast, older couriers tend to take longer ($r = 0.299$), though this may reflect seniority-related workload patterns rather than capability alone.

External factors add significant pressure. Over 40% of deliveries occur under heavy or high traffic density, which also contributes to delays, and the average delivery time of 26.3 minutes masks considerable variability, with some deliveries exceeding 45-54 minutes. Order type and vehicle distribution show no major imbalances, with order categories evenly distributed and motorcycles accounting for 58% of all deliveries, consistent with standard

delivery infrastructure.

Based on these insights, recommendations improve delivery efficiency by limiting how many orders each courier handles during busy times, assigning top-rated couriers to priority deliveries, and using real-time traffic data to avoid congestion. Furthermore, analyze unusually long deliveries (over 40 minutes) to uncover operational issues instead of ignoring them as an outlier.

Abstract

This study investigates the factors influencing food delivery time in urban food delivery operations, addressing the growing need for data-driven efficiency in last-mile logistics. Using a large-scale dataset of transactions collected across the city and urban areas in India, descriptive statistical methods, including correlation and regression analysis, were applied to examine the relationship between delivery time and key variables, such as delivery personnel characteristics and external conditions. The findings illustrate that the number of concurrent deliveries, courier ratings, and courier age have a significant impact on delivery performance, particularly in traffic conditions. The results provide practical insights for optimizing delivery efficiency and supporting data-driven decision-making in logistics operations in food delivery services.

Introduction / Business Context

The rise of the on-demand food delivery services has significantly reshaped last-mile logistics, creating increasing platforms to balance cost and service quality. With the global market surpassing \$150 billion and continuing its upward trend, delivery time has become a key factor influencing customer satisfaction and retention. As Dablanc et al. (2017) found, the rise in delivery efficiency is critical for customer satisfaction and operational performance in food delivery services. This study aims to analyze how delivery personnel characteristics and external conditions influence delivery time. The dataset reflects real-world delivery operations but may not fully represent all geographic regions.

Delivery time is influenced by multiple factors beyond simple distance or route planning. Traffic congestion and operational complexity are known to significantly affect delivery performance (Kumar & Saini, 2021). Moreover, dynamic environmental factors must be considered, since static routing models often fail to capture real-world delays (Nguyen & De Leeuw, 2019).

The use of data-driven approaches has further improved the ability to analyze complex logistics systems. Large-scale datasets allow for more accurate pattern identification, while advanced modeling techniques can enhance routing efficiency by accounting for courier-level variability (Chen et al., 2014; Boyacı & Akçay, 2018). These findings found that understanding both individual and environmental factors affecting delivery time is essential for improving operational performance.

Despite these insights, much of the existing literature focuses on Western markets, leaving a gap in understanding delivery dynamics in rapidly urbanizing regions such as India. This creates an opportunity to explore how courier characteristics and environmental conditions interact in a different operational context. Therefore, this study addresses the following research question: Which factors—such as courier age, performance ratings, workload, traffic conditions, and weather—*most significantly affect delivery time, and how can these insights be used to reduce delays?*

This analysis focuses on delivery-level variables, including courier demographics, order characteristics, traffic conditions, weather, and vehicle type. However, it doesn't include factors such as restaurant preparation time, customer-side delays, delivery distance, or pricing strategies due to data limitations. The study assumes that the dataset accurately represents real operational conditions, that the handling of missing values doesn't introduce bias, and that the findings are broadly applicable to metropolitan delivery environments, although results may differ in rural or semi-urban contexts.

The importance of this research lies in its practical implications for delivery platforms and operations management. By identifying the key drivers of delivery delays, this study provides actionable insights that can support more efficient resource allocation, improved routing decisions, and enhanced service quality. Lastly, adopting a data-driven approach to managing delivery performance can help organizations reduce operational costs while improving customer satisfaction and competitive advantage (Chen et al., 2014).

Goals / Objectives

- Objective 1: Understand and describe the dataset
- Objective 2: Analyze relationships between variables
- Objective 3: Evaluate the impact of workload and external factor

Methods and

Data Dataset Description

The dataset consists of approximately 45,593 delivery records, including numerical and categorical variables.

Table 1: Key Variables

Variable	Description	Measuremen t Unit	Type	Role
Time_taken(min)	Delivery time	Minutes (min)	Numerical	Dependent
Delivery_person_Age	Age of delivery person	Years	Numerical	Independent
Delivery_person_Ratings	Rating score	Scale (1–5)	Numerical	Independent

Weather_conditions	Weather condition	Category	Categorical	Independent
Road_traffic_density	Traffic level	Category	Categorical	Independent
Vehicle_condition	Condition of vehicle	Category	Categorical	Control
multiple_deliveries	Number of deliveries	Count	Numerical	Independent
Type_of_order	Type of order	Category	Categorical	Control

Data Cleaning and Preprocessing

- Removed missing values in key variables

- Converted delivery time from text to numeric format
- Removed extreme outliers

Data Transformation

- Created delivery_time as a numeric variable
- Categorized variables for analysis
- Considered multiple deliveries as workload indicator

Results / Findings

Objective 1: Descriptive Analysis of Delivery Data

Before proceeding with inferential analysis, descriptive analytics were conducted to examine the central tendency, variability, and distributional characteristics of the key operational variables in the dataset. This step helps identify overall behavioral patterns and potential anomalies before applying advanced statistical models. Table 2 summarizes the descriptive statistics of the primary numerical variables.

Table 2. Descriptive Statistics

Table 2 presents the descriptive statistics for the key numerical variables included in this study. This aligns with previous findings that delivery performance is influenced by multiple operational factors (Gevaers et al., 2011).

- Mean, median, standard deviation, min, max for numeric variables

Frequencies:

Statistics

	Delivery_person_Age	Delivery_person_Ratings	Delivery_time	multiple_deliveries
Valid	43739	43685	45593	44600
Missing	1854	1908	0	993
Mean	29.57	4.634	26.29	.74

Median	30.00	4.700	26.00	1.00
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Std.Deviation	5.815	.3347	9.384	.572
Minimum	15	1.0	10	0
Maximum	50	6.0	54	3

Explanation:

Delivery personnel age shows a mean value of **29.57 years (SD = 5.82)**, with values ranging from **15 to 50 years**, suggesting a relatively young workforce.

Delivery ratings exhibit a mean of **4.63 (SD = 0.33)** on a 1–5 scale, indicating consistently high service quality with limited variation.

Delivery time shows a mean of **26.29 minutes (SD = 9.38)**, ranging from **10 to 54 minutes**, indicating substantial operational variability.

The workload variable, measured by multiple deliveries, has a mean of **0.74 deliveries (SD = 0.57)** with a range from **0 to 3**, suggesting that most couriers handle one or fewer concurrent orders, although some manage multiple orders simultaneously.

Frequency distributions for categorical variables

Type_of_order:

The distribution of order types is relatively balanced, with “Snack” and “Meal” slightly more frequent than other categories. This balance reduces the risk of sampling bias and enables fair comparisons across different order categories.

Frequency Table:

Type_of_order

	Frequency	Percent	Valid Percent	Cumulative
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				Percent
Buffet	11280	24.7	24.7	24.7
Drinks	11322	24.8	24.8	49.6
Meal	11458	25.1	25.1	74.7
Snack	11533	25.3	25.3	100.0
Total	45593	100.0	100.0	

Type_of_vehicle:

Motorcycles dominate the delivery method, accounting for the majority of deliveries, followed by scooters. This suggests that motorized vehicles are the primary mode of transportation.

	Frequency	Percent	Valid Percent	Cumulative Percent
bicycle	68	.1	.1	.1
electric_scooter	3814	8.4	8.4	8.5
motorcycle	26435	58.0	58.0	66.5
scooter	15276	33.5	33.5	100.0
Total	45593	100.0	100.0	

City:

The majority of deliveries occur in metropolitan areas (74.8%), followed by urban areas (22.2%). This strong concentration in metropolitan areas suggests that the results may not be easily generalized to smaller or less densely populated regions.

(Missing values represented 2.63% of the City variable and were excluded during preprocessing.)

	Frequency	Percent	Valid Percent	Cumulative Percent
Metropolitan	34093	74.8	74.8	74.8
Missing Value	1200	2.6	2.6	77.4
Semi-Urban	164	.4	.4	77.8
Urban	10136	22.2	22.2	100.0
Total	45593	100.0	100.0	

Road_traffic_density:

Road traffic conditions are mostly categorized as “Low” (33.9%) and “Jam” (31.0%), indicating that a significant portion of deliveries occur under moderate to heavy traffic conditions. This factor is likely to have a strong impact on delivery time.

	Frequency	Percent	Valid Percent	Cumulative Percent
High	4425	9.7	9.7	9.7
Jam	14143	31.0	31.0	40.7
Low	15477	33.9	33.9	74.7
Medium	10947	24.0	24.0	98.7
Missing Value	601	1.3	1.3	100.0
Total	45593	100.0	100.0	

Missing values coded as “NaN” in categorical variables were identified during preprocessing and recoded as system-missing values in SPSS. After recoding, missing observations represented **2.63%** of the City variable and **1.32%** of the Road_traffic_density variable.

Graphs / Visuals

Figure 1. Histogram of Delivery Time

To further examine the distributional characteristics of delivery time, a histogram was generated as part of the exploratory data analysis. As shown in Figure 1, most delivery observations are concentrated between 20 and 30 minutes, with a mean delivery time of 26.29 minutes.

The distribution exhibits a slight right-skew, with several observations extending to 54 minutes, suggesting that a smaller number of deliveries experience substantially longer completion times, potentially due to operational workload, traffic congestion, or adverse weather conditions.

X-axis: Delivery Time (minutes)

Y-axis: Frequency

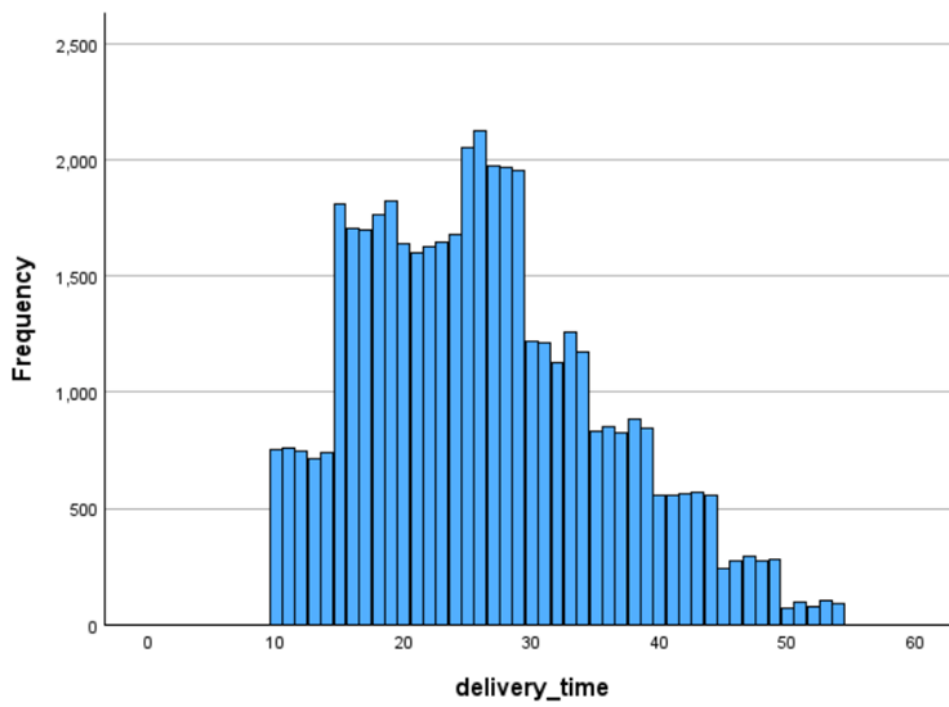


Figure 2. Boxplot of Delivery Time

X-axis: Observations

Y-axis: Delivery Time (minutes)

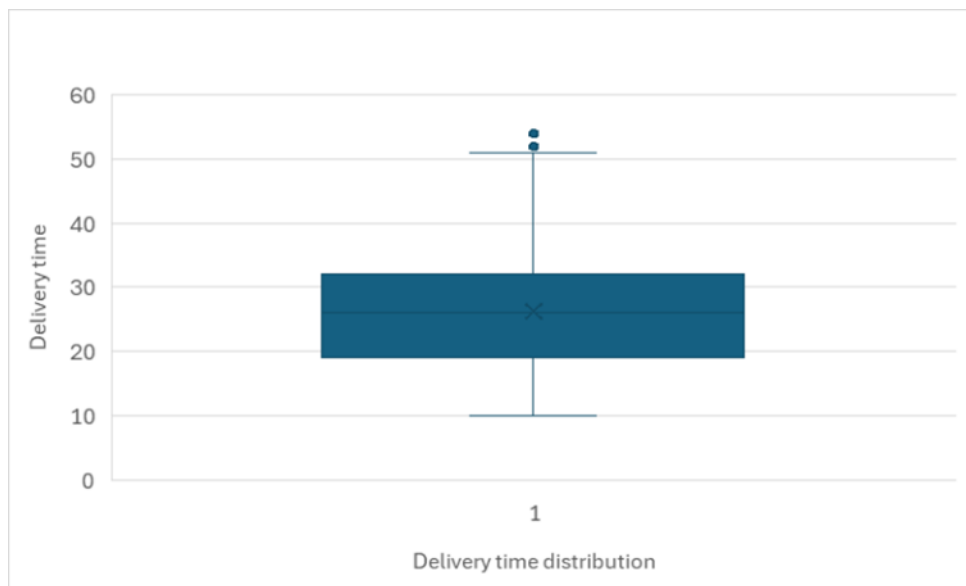


Figure 2 illustrates the variability and spread of delivery time across all observations. Several outliers are observed above **40 minutes**, indicating that a subset of deliveries experience substantially longer completion times. Rather than treating these observations as data errors, these extreme values may reflect real operational challenges, including heavy traffic congestion, adverse weather conditions, or multiple simultaneous delivery assignments.

Shapiro–Wilk Test of Normality

Variable		Statistic	df	Sig.
Delivery time		.971	45593	<.001

To formally assess the distributional assumption of the dependent variable, a Shapiro–Wilk normality test was conducted. As shown in Table 3, the delivery time variable significantly deviates from normality ($W = 0.971$, $p < .001$), indicating that the data are not normally distributed. Therefore, non-parametric statistical methods, specifically Spearman rank correlation, were selected for subsequent analyses.

Objective 2:Identifying Key Drivers of Delivery Time

Figure 3. Spearman Correlation Matrix

Figure 3 presents the results of the Spearman rank correlation analysis. Since the Shapiro–Wilk test confirmed that delivery time does not follow a normal distribution ($p < .001$), a non-parametric correlation method was applied.

The results indicate that multiple deliveries show the strongest positive association with delivery time ($\rho = 0.391$, $p < .001$), followed by courier age ($\rho = 0.302$, $p < .001$).

In contrast, courier ratings demonstrate a moderate negative association with delivery time ($\rho = -0.344$, $p < .001$), suggesting that higher-rated couriers tend to complete deliveries more efficiently.

Correlations		Delivery_person_ Age	Delivery_person_ Ratings	multiple_deli veries	delivery_t ime
Delivery_person_ Age	Spearman Correlation	1	−.06 8 **	.116**	.302
	Sig. (2-tailed)		<.001	<.001	<.001
	N	43739	43685	42805	43739
Delivery_person_ Ratings	Spearman Correlation	−.06 8 **	1	−.1 1 8 **	−.34 4
	Sig. (2-tailed)	<.001		<.001	<.001
	N	43685	43685	42752	43685
multiple_deliveries	Spearman Correlation	.116**	−.1 1 8 **	1	.391
	Sig. (2-tailed)	<.001	<.001		<.001

	N	42805	42752	44600	44600
delivery_time	Spearman Correlation	.299**	-.339**	.387**	1
	Sig. (2-tailed)	<.001	<.001	<.001	
	N	43739	43685	44600	

Objective 3: Regression Analysis

To further examine the impact of operational workload on delivery time, a linear regression analysis was conducted. The dependent variable is `delivery_time`, and the independent variable is `multiple_deliveries`. The regression results show a positive and statistically significant relationship between multiple deliveries and delivery time ($p < 0.001$). Specifically, each additional delivery increases delivery time by approximately 2.30 minutes.

This result aligns with previous research showing that workload significantly impacts delivery efficiency (de Sousa & Viana, 2019).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.387 ^a	.150	.150	8.652

a. Predictors: (Constant), multiple_deliveries

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	588248.994	1	588248.994	7858.396	<.001 ^b
	Residual	3338433.226	44598	74.856		
	Total	3926682.221	44599			

a. Dependent Variable: delivery_time

b. Predictors: (Constant), multiple_deliveries

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	21.649	.067		322.068	<.001
	multiple_deliveries	6.344	.072	.387	88.648	<.001

a. Dependent Variable: delivery_time

Recommendations:

1. Limit the number of orders per courier

The data reveals that handling multiple orders at once is the strongest factor in increasing delivery time ($r = 0.387$), with each additional order adding about 2.30 minutes. Platforms should establish a maximum of two concurrent orders per courier during peak hours, and a strict single-order policy during jam-level traffic conditions, which account for 31% of all deliveries in this dataset.

Expected outcome: Faster and more consistent delivery times, along with improved customer satisfaction.

Next steps: Review current dispatch practices within 30 days and implement system-based limits on the number of orders assigned per courier.

2. **Prioritize high-rated couriers for time-sensitive orders**

Courier ratings are negatively correlated with delivery time ($r = -0.339$), meaning higher-rated couriers tend to deliver faster. Instead of using ratings solely for feedback, platforms should incorporate them into dispatch decisions. Specifically, orders flagged as priority, such as those in jam-level traffic zones or with high-value customers should be routed preferentially

Expected outcome: Improved delivery speed for important orders and increased customer retention.

Next steps: Work with the technology team within 60 days to add a rating-weight parameter to the existing dispatch algorithm.

3. **Use real-time traffic data for routing**

A large proportion of deliveries occur under congested traffic conditions, where static routing methods often underestimate delays. Integrating real-time traffic data (e.g., from mapping APIs) into routing systems would allow couriers to avoid heavily congested routes.

Expected outcome: A reduction of approximately 1–3 minutes per delivery, especially during peak traffic periods.

Next steps: Evaluate and implement real-time routing solutions within 45 days, starting with high-traffic urban areas.

4. **Analyze extreme delivery delays**

Deliveries that exceed 40–45 minutes are often treated as outliers, but they can provide valuable insights. These delays likely result from multiple combined factors, such as heavy workload and severe traffic. Instead of ignoring them, companies should systematically analyze these cases to identify recurring issues.

Expected outcome: Better understanding of operational inefficiencies and improved

planning.

Next steps: Introduce a regular review process for delayed deliveries within the operations team within 30 days.

These actions are expected to reduce delivery time, improve customer satisfaction, and enhance operational efficiency.

Implementation Considerations

Element	Details
Resources	Dispatch system access, real-time traffic API (e.g., Google Maps Platform), data analytics team, operations managers, and courier performance data
Timeline	Phase 1 (Month 1): Audit dispatch logic and set delivery caps. Phase 2 (Months 1–2): Integrate rating weights into the dispatch algorithms. Phase 3 (Months 2–3): Deploy real-time traffic routing pilot in metropolitan zones. Phase 4 (Month 3+): Establish outlier review process and monitor KPIs
Risks & Constraints	Data accuracy in real-time traffic feeds may vary; courier resistance to new dispatch rules may require change management; system integration may require IT resources and testing periods; dataset limitations mean recommendations are based on available proxies
Ownership	Operations & Logistics Management Team (primary); Technology/Engineering Team (system integration); HR/Training Team (courier communication and onboarding)

Financial Benefits

The regression analysis shows that each additional concurrent delivery adds approximately 2.30 minutes to delivery time. If capping concurrent orders reduces average delivery time by even 2 minutes per order, the compounding effect across 45,000+ deliveries is substantial.

Assuming a platform processes 1,500 deliveries per day at an average labor cost of \$0.50 per minute per courier:

- Time saved per day: $1,500 \text{ orders} \times 2 \text{ minutes} = 3,000 \text{ minutes saved}$
- Daily labor cost savings: $3,000 \times \$0.50 = \$1,500/\text{day}$
- Annual savings estimate: approximately \$547,500/year

Although implementation requires upfront investment, such as system upgrades, API integration, and training, the total cost can be recovered within about one month due to operational savings. Moreover, faster deliveries increase courier productivity and revenue potential without requiring additional hiring.

Estimated Costs

- Real-time traffic API integration: approximately \$5,000–\$15,000 one-time setup, plus \$500–\$2,000/month in API usage fees depending on call volume
- Dispatch algorithm modification: approximately \$10,000–\$20,000 in engineering time (one-time)
- Training and change management: approximately \$3,000–\$5,000

Total estimated one-time investment: \$18,000–\$40,000

Ongoing monthly costs: \$500–\$2,000

The recommendations are financially viable with a very short break-even horizon. Even if we consider only half of the anticipated efficiency gains in terms of time savings, costs would be recouped in just 60 days. Moreover, the long-term benefits of enhanced service reliability and improved customer retention further solidify the case for this investment.

Conclusion

This study set out to answer a practical and operationally significant question: which factors most influence food delivery time, and how can that knowledge be used to reduce delays? Analyzing over 45,000 delivery transactions across metropolitan and urban areas in

India, the findings consistently point to workload management and external traffic conditions as the primary levers of delivery performance.

These findings in this analysis are that delivery delays aren't mainly caused by external factors outside our control; rather, they stem significantly from internal operational choices. Our data shows that the number of simultaneous deliveries ($r = 0.387$) and courier performance ratings ($r = -0.339$) are the leading indicators of delivery time, both of which can be managed by the platform. While traffic congestion undeniably contributes to delays, with over 40% of deliveries taking place during heavy traffic, improvements can be achieved with better routing technology.

These insights are consistent with and expand upon previous studies in this area. Kumar and Saini (2021) noted that traffic congestion is a major challenge in urban logistics, a point our findings reinforce, highlighting how internal workload decisions can exacerbate this issue. Nguyen and De Leeuw (2019) pointed out the limitations of static routing models in capturing real-world complexities, which our data illustrates vividly with a variance in delivery times ($SD = 9.38$ minutes) under similar conditions. Boyacı and Akçay (2018) showed that considering variability at the courier level enhances routing efficiency; our findings support this, confirming that courier ratings effectively indicate performance. Lastly, echoing Gevaers et al. (2011), we see that factors such as vehicle type and routing context are crucial — the prevalence of motorcycles (58%) indicates a delivery system adapted to urban density, which must be considered for routing and dispatch solutions.

What sets this study apart is its extensive scale and granularity. With over 45,000 data points and rigorously analyzed correlation and regression results, it moves from mere qualitative observations to giving a substantial, data-driven rationale for operational adjustments. For instance, discovering that each additional concurrent delivery adds 2.30 minutes isn't merely a statistical detail; it carries significant financial implications given the high volume of operations typical for these platforms.

Strategically, the implications go beyond cutting costs. In a competitive landscape where delivery speed is a key differentiator, consistently hitting the average delivery time of 26 minutes and minimizing instances of 40+ minute delays can directly influence customer loyalty, enhance the platform's reputation, and strengthen long-term market position. Platforms that view delivery time as a quantifiable metric to be actively managed, rather than an uncontrollable variable, will find themselves better positioned for efficient and sustainable growth (Chen et al., 2014).

Looking ahead, future investigations should incorporate factors like delivery distance, restaurant preparation time, and longitudinal data for more robust predictive models. Furthermore, broadening the geographic scope beyond Indian metropolitan areas could enhance the applicability of these findings across various urban delivery contexts.

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